LEONARDO V. CASTORINA

leonardo.castorina@outlook.com | REDACTED ONLINE | DrLeucine | in/leonardo-castorina

EDUCATION

The University of Edinburgh

- PhD in Machine Learning for Protein Design — Sep 2021 - Apr 2025
- MScR in Artificial Intelligence — Sep 2020 - Sep 2021
- BSc (Hons) in Biochemistry — Sep 2016 - May 2020

EXPERIENCE

Apheris – *Senior ML Engineer* — Jun 2026 - *Present*

AstraZeneca – *Senior Research Scientist* — Apr 2025 - May 2026

- Performed *de novo* design and optimisation on 5 targets for affinity, cross-reactivity, developability.
- Engineered biochemically interpretable statistical models for multispecific antibody pairing.
- Designed geometric representations for rapid off-target and interaction screening.

NEC OncoImmunity – *ML Consultant* — Nov 2024 - Feb 2025

- Developed vaccine design pipeline yielding potent candidates for cancer and infectious diseases.

Microsoft Research – *ML Consultant* — Nov 2023 - Apr 2024

- Developed statistical framework to analyse sequence variance in the context of TCR repertoires.
- Mapped sequence polymorphisms to 3D structural features to disentangle interaction drivers.

Microsoft Research – *Research Scientist Intern* — Jun 2023 - Sep 2023

- Mined 30K TCR repertoires with MHC and peptide data to identify key interaction patterns.

NEC Labs Europe – *Research Scientist Intern & ML Consultant* — Oct 2022 - Apr 2023

- Developed GNN models integrating 3D and biological features for TCR-pMHC binding prediction.

SELECTED PUBLICATIONS

From Atoms to Fragments: A Coarse Representation for Efficient and Functional Protein Design

Leonardo V. Castorina, Christopher W. Wood, Kartic Subr (*Bioinformatics*, 2026)

HLA Alleles Shape Distinct Biases in the Usage Preferences of TCR Vβ segments

Leonardo V. Castorina, Matthew T. Noakes, [...], H. Jabran Zahid (*Frontiers in Immunology*, 2026)

TIMED-Design: Flexible and Accessible Protein Sequence Design with CNNs

Leonardo V. Castorina, Suleyman Mert Ünal, Kartic Subr, Christopher W. Wood (*PEDS*, 2024)

PROJECTS

Tessera *Fragment-Based Protein Representations* — GitHub

- Engineered fully-vectorised Python library for coarse-grained functional protein representations.
- Functional conditioning for RFDiffusion, up to 68× faster search, improved functional clustering.

TIMED-Design *Deep Learning for Protein Design* — GitHub | Try it out

- Developed, reproduced, and benchmarked CNN models for protein design and an easy-to-use UI.

Aposteriori *Protein Structures Voxelisation* — GitHub

SKILLS

Tools: AlphaFold, Boltz, Chai, ESM, Git, Jupyter, ProteinMPNN, PyMOL, Python, Streamlit.
Libraries: Keras, NumPy, Pandas, PyTorch, SciKit-Learn, TensorFlow.
Machine Learning: CNNs, GANs, GNNs, Geometric DL, Structural Bioinformatics, Transformers, VAEs.